

Automated Painting Rover (APR) MSME

Design Review Research Document

*Q1 Market Landscape · Q2 Labour Cost Analysis · Q3 Finalized
Technology Stack*

**Prepared for: Formal Design Review Submission
Document Date: 1 September 2026**

Anurag University

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Automatic Painting Machine Robotic Systems Robotic Arm Welding Robots Spray Painting Equipment Paint Spray Gun Paint Sprayers Premium Paint Brushes

Automatic Painting Machine Robotic Systems Robotic Arm Below ₹10,00,000 Pune Chennai Gurugram Maharashtra v Welding Robots Spray Painting Equipment

Aluminium Pick System, For Pick, Computerized

₹ 18,50,000

Contact Supplier

System Control	Computerized
Material	Aluminium
Usage/Application	Pick
Light Sensor	Yes

Robotic Painting Systems

₹ 15,00,000

Contact Supplier

Robot Type	Articulated
No Of Axes	6 Axis
Payload Capacity	10 kg
Reach	2000 mm

Painting Robotic Systems, For Industrial

₹ 6,50,000

Contact Supplier

Country of Origin	Made in India
Surface Finish	Power Coated
Usage/Application	Industrial
Type	Spray Gun

Industrial Painting Robot

₹ 7,00,000

Contact Supplier

Color packaging	Standard Box
Handle Material	Mild Steel
Usage/Application	Industrial

Spray Gun Robotic Painting Systems

₹ 10,00,000

Contact Supplier

Type	Spray Gun
Country of Origin	Made in India

our painting robot is an automated system designed to apply paint or coatings with

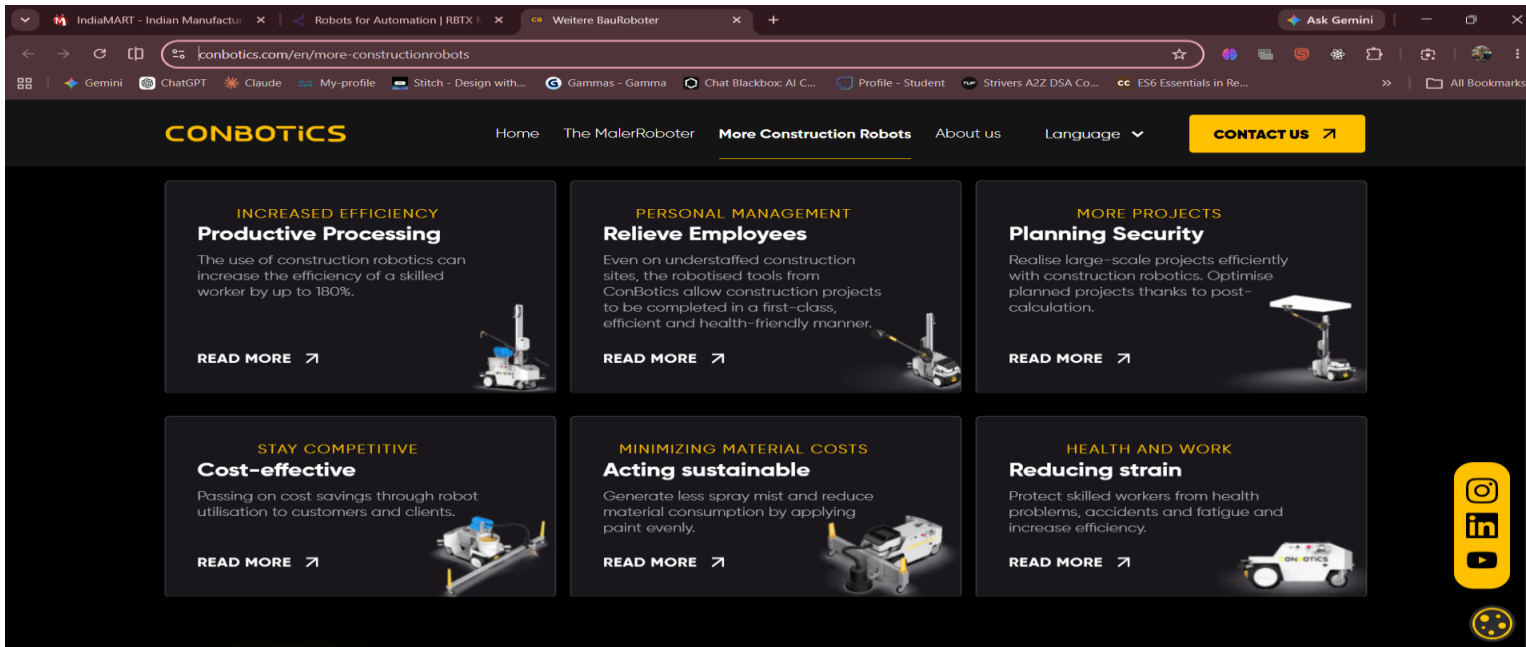
IndiaMART - Indian Manufactur x Robots for Automation | RBTX x Weitere BauRoboter x + Ask Gemini

rbtx/en-IN/components/robots

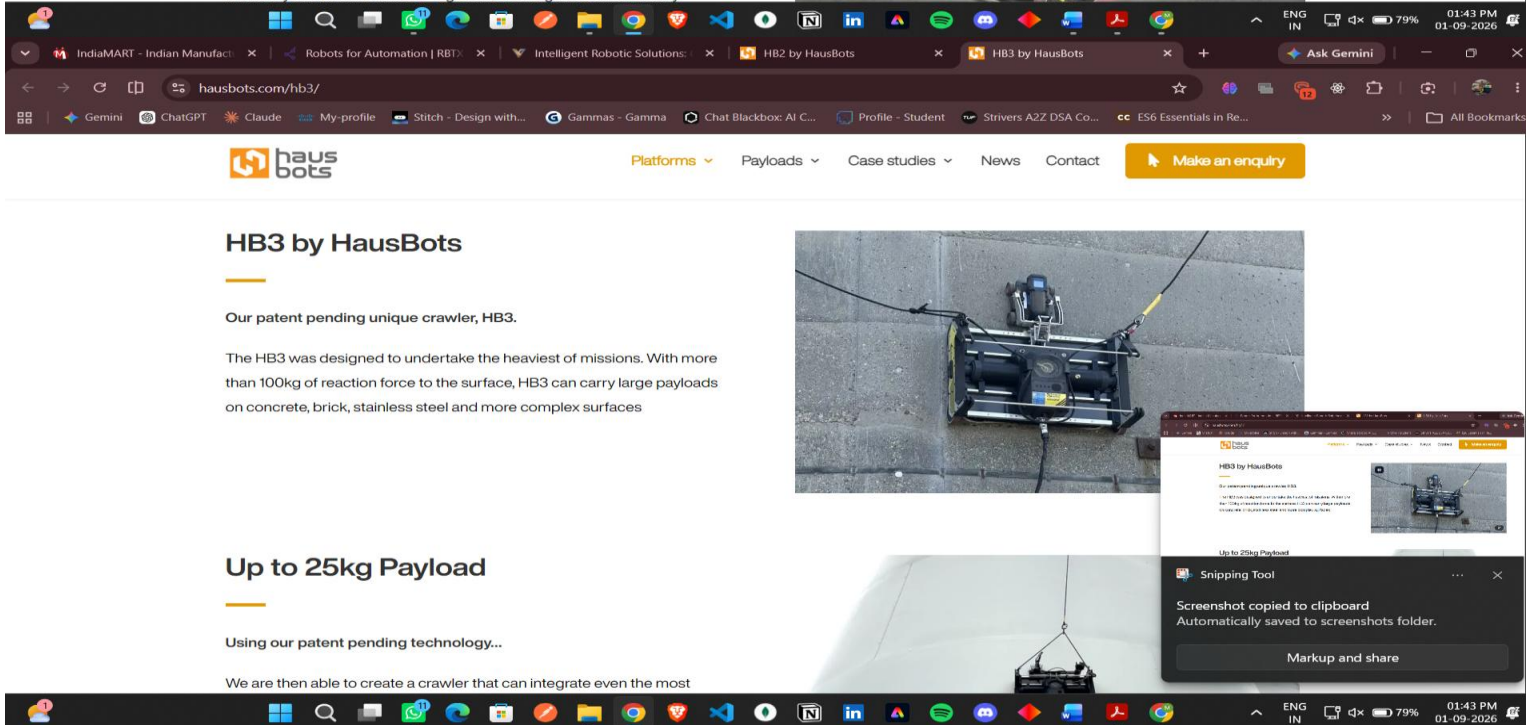
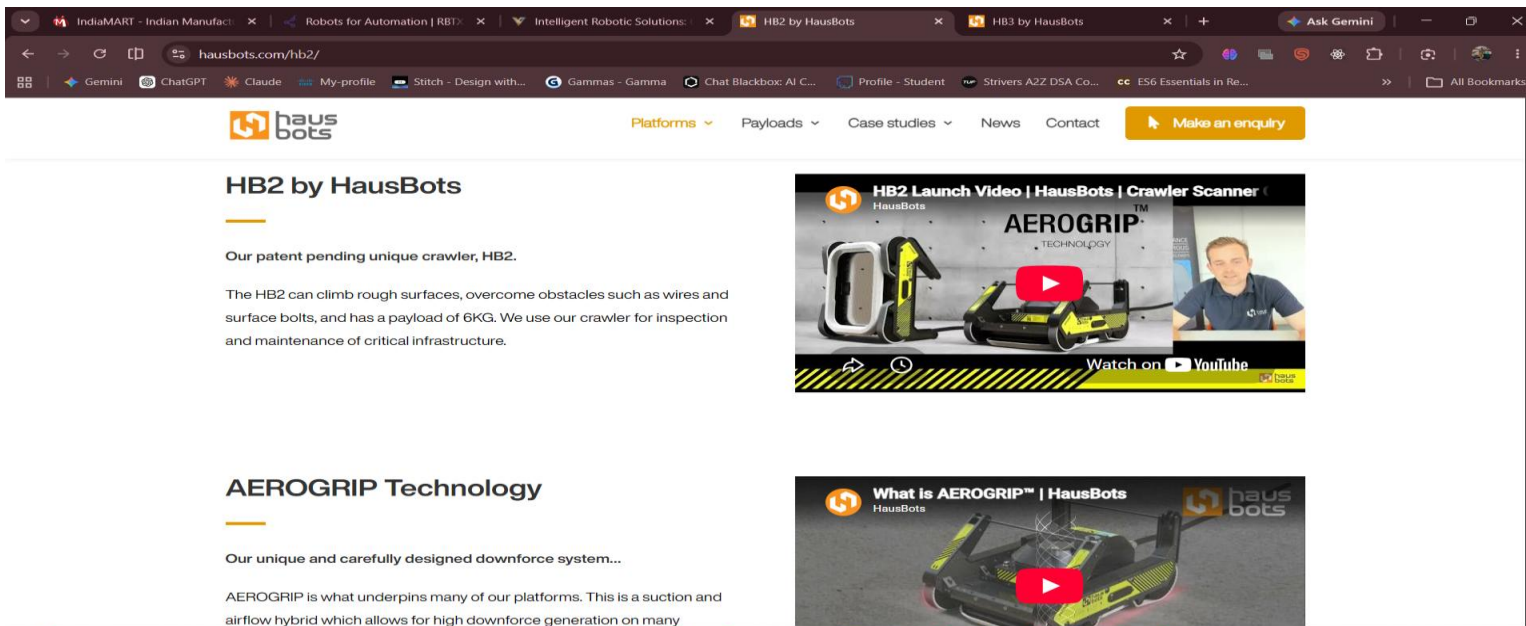
Gemini ChatGPT Claude My-profile Stitch - Design with... Gamas - Gamma Chat Blackbox: AI C... Profile - Student Strivers A2Z DSA Co... ES6 Essentials in Re... All Bookmarks

<https://www.conbotics.com/en/more-constructionrobots>

A screenshot of the CONBOTICS website. The header features the CONBOTICS logo and navigation links: Home, The MalarRoboter, More Construction Robots, About us, and Language. A yellow 'CONTACT US' button is in the top right. The main content area has a dark background with the text 'BauRoboter. Easy. Use.' and 'For specialists on construction sites buy BauRoboter.' Below this is another yellow 'CONTACT US' button. To the right, a large yellow circle highlights five different BauRoboter models, each with a label: 'Walls & Ceilings', 'Sanding', 'Coating', 'Floor', and 'CONBOTICS' (the central robot). A vertical sidebar on the far right contains social media icons for Instagram, LinkedIn, and YouTube, along with a small circular icon at the bottom.



HausBots <https://hausbots.com/>



2. Q1 — Is an Automated Painting Rover Available Today?

Finding: Yes, but only as a B2B industrial product — not on any consumer marketplace. -> ****Available in UK.**

2.1 Consumer marketplaces

Searches on Amazon and Flipkart for "wall painting robot" return only decorative canvases, children's toys, and the Scribit wall-drawing plotter (a pen device, not a painting machine). No automated wall-painting rover is listed for retail sale.

2.2 Indian competitors — must be acknowledged explicitly

Company	Location	Product	Key verified facts
Vividobots	Chennai	Wiperque (façade cleaning, live) / Vibrantque-XT (exterior painting, in testing)	Markets itself as India's first autonomous exterior painting machine; 40 L paint canister; robotic arm; ₹1.85 cr+ raised across funding rounds; incubated at Anna Incubation Centre.
Pace Robotics	Bengaluru	Centa Painter	Interior wall/ceiling putty + paint robot; LiDAR + camera + NVIDIA processor; ~10× speed, ~80% less manpower claimed.
Endless Robotics	Bengaluru (Hyderabad projects)	Walt	Interior painting robot; completed projects with Meenakshi Group and Kalpataru Group in Hyderabad; report dates to 2019 — re-verify current status before citing.
Eternal Robotics	India-origin	MYRO	Indoor painting robot driven from a tablet/floor plan; not fully autonomous.

2.3 International commercial products

Product / Company	Country	What it does
Okibo (EG7)	Israel	Fully autonomous interior painting; ~1,000 sq ft/hr; \$7.85M raised (Mar 2024).
Legend Robot	China	Latex spray robots, up to 6.2 m reach; >95% coverage claimed; 2,400–3,000 m²/day.
PaintJet	USA	Lift-mounted robotic spray system for building/warehouse exteriors — automates spraying only, not access.
Robosurf	Italy	Autonomous mobile robot painters for construction/industrial/naval sectors.
Atombrush	Singapore	Robotics for automating building painting, addressing labour shortages.

2.4 Wall-climbing robots — APR's exact category

HausBots (UK, University of Warwick spin-out) is the closest reference point. Its AEROGRIP system uses airflow-generated pressure differential rather than a maintained suction seal, which is why it can climb rough, dusty, non-smooth surfaces.

Model	Verified figures
HB1 (earlier)	4 rubber-tyred wheels; tethered 110V; 6 kg payload; ~42 kgf (≈412 N) force; 30 m range.
HB2/HB3 (current)	AEROGRIP adhesion; ~350 N force; works on steel, concrete, brick, composites; floor-tethered for continuous power.

Why this matters for APR: HausBots HB2 is a commercially deployed robot generating ~350 N of adhesion. APR's own requirement targets ≥500 N for a rover roughly twice that weight-adjusted class. The HausBots figure is independent evidence that airflow-based adhesion in the 350–450 N range is physically achievable — it makes APR's target look researched, not invented.

Historical precedent also exists: the Kumagai-Gumi wall-climbing robot (Japan) used vacuum suction for shot-blasting and painting; Shimizu OSR-I and the Taisei painting robot are further precedents. Academically, VIT Chennai published PAINTBOT-EW (2016), an exterior wall painting robot with a suspension system.

2.5 Price landscape

Segment	Reported price range
Basic (manual control, simple spray)	\$3,000 – \$7,000
Semi-automatic (motorised, partial automation)	\$8,000 – \$15,000
Fully automated (AI-assisted, advanced sensors)	\$16,000 – \$40,000+
Business "sweet spot" (market guide)	\$30,000 – \$100,000
Overall 2026 market range	\$5,000 – \$200,000+

These are vendor/market-guide figures, not audited prices — flagged as indicative in the source report.

3. Q2 — Labour Cost to Paint a Wall in Telangana (FINAL)

FINAL ADOPTED FIGURE: ₹750 – ₹1,200 per day for a skilled painter in Telangana.

This range is now fixed as the headline labour-cost figure used throughout the APR project documentation and cost-competitiveness argument, replacing the wider set of estimates in the original research pass. It sits consistent with — and slightly tightens — the underlying research, summarized below for traceability.

3.1 How this figure is supported by the underlying research

Basis	Reported figure	Source
Statutory minimum wage, skilled worker, Zone I (Hyderabad), from 1 June 2026	₹712/day	Telangana Labour Dept. notification, 30 May 2026
Market rate, skilled painter (India-wide, 2026)	₹800 – ₹1,300/day	Solve24 painter salary survey, 2026
Market rate, painting contractor	₹1,500 – ₹2,200+/day	Solve24 painter salary survey, 2026
Comparable city (Chennai) skilled painter rate	₹800 – ₹1,100/day	AapkaPainter Chennai guide, 2026

₹750–₹1,200/day sits just above the statutory floor and comfortably inside the market-rate band reported for skilled painters, while staying below full contractor pricing — a defensible, round figure for a design-review document.

3.2 Per-wall cost context (8 ft × 8 ft = 64 sq ft example)

Using the finalized daily rate and the productivity assumption that a small 64 sq ft wall takes roughly one-third of a working day, but that painters typically charge a half-day to full-day minimum for a job this small:

- Strict pro-rata (1/3 day): ₹750–1,200 × 1/3 ≈ ₹250 – ₹400
- Practical minimum billed (half day to full day): ₹375 – ₹1,200
- All-in with materials (interior, standard/mid-range, per earlier ₹/sq ft research): ₹1,280 – ₹1,920 for 64 sq ft

4. Q3 — Finalized Technology Stack (FINAL)

The new research ("APR — Research Findings with Sources") was compared against the existing software architecture, requirements document, and project report. Conclusion: the existing architecture is not replaced. It is upgraded in three places and confirmed everywhere else.

Governing principle: use AI where uncertainty exists (perception); use deterministic algorithms where correctness and safety matter (navigation, control, safety).

4.1 What changed vs. the original plan

Area	Original plan	Final decision	Why
Onboard compute	Raspberry Pi	NVIDIA Jetson Orin Nano Super	Up to 67 INT8 TOPS, 8GB LPDDR5, 7–25W configurable — required to run YOLO + SLAM + Nav2 concurrently on one board.
Perception model	YOLOv8	YOLO26 (primary), YOLO11 (fallback)	YOLOv8 is two generations old by 2026. YOLO26 is NMS-free and up to 43% faster CPU inference than YOLO11-N; YOLO11 is the stable production fallback.
Dashboard role	MERN as control layer (implied)	MERN/React as Digital Twin + Mission Control + Telemetry only	The dashboard must never write directly to motors, spray, or vacuum — it sits above ROS 2, not inside the control loop.

4.2 Decision pipeline

Camera feed → YOLO26 detector (finds windows, pipes, edges) → Coverage planner (Boustrophedon decomposition) → PID standoff control (holds distance, sets spray rate) → Sensor fusion + threshold (pressure/IMU, grip health). Detection feeds the planner only; control and safety loops stay classical and deterministic — AI output never reaches motors, spray valves, or vacuum control directly.

4.3 Subsystem-by-subsystem final decisions

Subsystem	Final choice	One-line justification
Operating system	Ubuntu 24.04 LTS	Recommended pairing with ROS 2 Jazzy + Gazebo Harmonic.
Middleware	ROS 2 Jazzy	Backbone for sensors, localization, navigation, planning, controllers, safety, hardware.
Simulation	Gazebo Harmonic	Direct ROS 2 integration via ros_gz.
Visualization	RViz2	Engineering/debug tool — distinct from the APR Command dashboard.
Navigation	Nav2	Handles costmaps, controllers, goal execution, recovery — not reimplemented in-house.
SLAM	SLAM Toolbox, with CAD as prior map	CAD supplies known wall geometry; sensors verify/localize rather than reconstructing the wall from scratch.
Sensor fusion	robot_localization / EKF	Fuses wheel encoders, IMU, odometry into robot pose.
Coverage planning	Boustrophedon Cellular Decomposition	Provably complete coverage; matches the ≥90% coverage requirement.
Adhesion	Active negative-pressure vacuum + compliant skirt	Only physically viable method on non-ferrous, porous concrete/brick/plaster; validated at 350–412 N by HausBots.
Locomotion	Tracks	Larger contact patch maintains the pressure differential and improves grip on rough plaster/weld lines.
Motion/spray control	Closed-loop PID	Deterministic, cheap, no training data needed; holds 200–300 mm standoff at ±20 mm tolerance.
Safety	Deterministic pressure + IMU threshold state machine	A safety trigger must be provable before a live demo — not statistical.

Subsystem	Final choice	One-line justification
Perception	YOLO26 (fallback YOLO11) + OpenCV	Current generation, NMS-free, edge-optimized; feeds obstacle costmap and dashboard only.
Compute (onboard)	NVIDIA Jetson Orin Nano Super	Runs ROS 2 + Nav2 + Perception + SLAM concurrently; same class of processor used by Pace Robotics' Centa Painter.
Low-level control	Separate microcontroller	If the Linux/Jetson side freezes, the microcontroller still enforces motor and safety I/O.
CAD input	DXF + Python parser (DWG deferred)	AutoCAD supplies geometry; it does not control the robot.
Web / digital twin	React + TypeScript + Three.js + Node.js + Express + MongoDB + Socket.IO	Digital Twin, Mission Control and Telemetry layer only — never writes directly to actuators.
Robot-to-dashboard link	ROS 2 → ROS-Web bridge → Node.js → Socket.IO → React	Keeps ROS 2 as the robot-side backbone and Socket.IO as the web-side transport.
Languages	Python (planning, CV, ML, tooling) + C++ (high-frequency control, hardware interfaces)	Matches responsibilities to performance needs.
Logging	rosviz	Standard ROS 2 experiment recording.
Version control	Git / GitHub	No change.

4.4 AI / Perception — phased rollout and licensing

Licensing flag for the review panel: YOLO26 ships under AGPL-3.0, or a paid Enterprise license for closed-source commercial use. Since APR has a stated commercialization objective, this must be a documented decision, not something discovered late. Raise it proactively — it signals commercial awareness.

- Phase 1 — No deep learning. Navigate using LiDAR + known CAD obstacles only.
- Phase 2 — Add OpenCV + camera (classical CV).
- Phase 3 — Add YOLO26 for prototype perception, once licensing is resolved.
- Phase 4 — Benchmark YOLO26 vs YOLO11 vs a commercially-clear alternative on APR's own dataset, scoring FPS, precision, recall, latency, power draw, false positives, obstacle detection distance, and licensing — then freeze the model.

AI output is confined to the obstacle costmap and operator dashboard. It never has write access to motors, spray valves, or vacuum control — this is the AI isolation principle already established for APR and it is reconfirmed, not changed, by this research pass.

4.5 Compute and hardware architecture

Layer	Hardware	Responsibility
Development	Engineer's laptop	Ubuntu 24.04, ROS 2 Jazzy, Gazebo, RViz, VS Code, Git.
Onboard (high level)	Jetson Orin Nano Super	ROS 2, Navigation, Perception, AI, sensor processing.
Onboard (low level)	Separate microcontroller (serial/CAN link to Jetson)	Motors, encoders, vacuum, spray, safety I/O — keeps functioning even if Linux freezes.

4.6 Removed from the plan

Removed item	Replacement / reason
Raspberry Pi as primary onboard computer	Jetson Orin Nano Super — insufficient AI headroom on Pi for concurrent YOLO + SLAM + Nav2.
YOLOv8	YOLO26 / YOLO11 — two generations newer, better edge performance.
Node-RED	Not needed for V1; revisit only if external IoT/cloud workflow integration is required later.
AI-controlled motors / end-to-end AI navigation	Never — violates the AI isolation and safety-determinism principles.
Custom SLAM / custom navigation framework	SLAM Toolbox + Nav2 — do not reimplement what already exists and is proven.
Reinforcement-learning navigation (initial)	Deferred — no completeness guarantee, needs training data the project doesn't have time to collect.
DWG as first-class CAD input	DXF first; DWG deferred.

4.7 V1 scope discipline

The first prototype targets a flat vertical wall of known geometry, with no complex obstacles, delivering autonomous coverage, basic localization, and safety. Complex façades, mid-mission obstacles such as windows and pipes, and severe weather are explicitly deferred to later phases (V2 known obstacles/no-paint zones, V3 unexpected obstacle detection, V4 dynamic replanning, V5 advanced AI) — consistent with the requirements document scoping already in place.

5. Full Source List

All URLs were accessed on 31 August 2026 as part of the underlying research pass. Figures quoted are as reported by these sources; vendor claims are identified as such in the text above.

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